

Ye Jenkins Pipeline kya karti hai?

Ye pipeline automatically:

1. Code download karti hai (Git se)
2. Maven se WAR file banati hai
3. Tomcat me deploy karti hai
4. Tomcat restart karti hai
5. Application check karti hai (chal rahi hai ya nahi)

Matlab: **Full CI + CD automation**

1. agent any

agent any

→ Pipeline kisi bhi available Jenkins machine (agent) par run ho sakti hai.

2. environment block

Yahan saare tools ke paths define kiye gaye hain:

JAVA_HOME

MAVEN_HOME

TOMCAT_HOME

TOMCAT_WEBAPPS_PATH

TOMCAT_SERVICE_NAME

APP_URL

Simple meaning:

- Java kaha installed hai
- Maven kaha installed hai
- Tomcat kaha installed hai
- WAR file kis folder me copy hogi
- Tomcat service ka naam
- Application ka URL

Taaki har stage me same values use ho saken.

3. Stage: Cleanup Workspace

deleteDir()

→ Purana build data delete karta hai

→ Fresh build ke liye clean folder

4. Stage: Clone Repository

git url: '...', branch: 'main'

→ GitHub se project code download karta hai

5. Stage: Build WAR File

mvn clean package -DskipTests

→ Maven se:

- old build delete
- new build create
- WAR file generate

File banegi:

target/project_3.war

6. Stage: Copy WAR to Tomcat

Yeh 3 kaam karta hai:

1. Old folder delete

rmdir project_3

2. Old WAR delete

del project_3.war

3. New WAR copy

copy target\project_3.war tomcat\webapps\

→ Fresh deployment ke liye

7. Stage: Ensure Tomcat Service Exists

sc query Tomcat9

→ Check karta hai Tomcat service installed hai ya nahi

Agar nahi:

service.bat install

8. Stage: Restart Tomcat

sc stop Tomcat9

sc start Tomcat9

→ Tomcat restart karta hai taaki new WAR load ho jaaye

9. Stage: Verify Deployment

Invoke-WebRequest http://localhost:8080/project_3

→ Application 3 baar try karta hai

- Agar open hui → SUCCESS
 - Nahi hui → FAIL
-

10. post section

always

Workspace clean

success

 DEPLOYMENT SUCCESSFUL

failure

 DEPLOYMENT FAILED